

*Explanation
of
Attributes and Attribute Categories
used in
Geologic Map Digital Database Files
produced by*

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Geologic Mapping Program**

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Introduction

Digital GIS data files of geologic maps produced by the Utah Geological Survey have three types of data: arcs, polygons, and point data. A special class of point data is annotations (text that actually appears on the map), which has two types: (1) names or information that are attributes of arcs, polygons, or points (such as the map unit label, name of a fault, or the dip angle of a strike and dip symbol), and (2) short strings of text placed on the map by the geologist to label, point out, or explain some feature, such as “breccia zone,” “disturbed area,” etc.

The following list provides a brief explanation of the primary attributes and attribute categories used on the arcs, polygons, and point data. This list applies to most geologic maps; no map has every type of attribute. A few maps have additional rarely used attributes not explained below, but which are explained on the explanation plate accompanying such maps. The actual list of all attribute possibilities is very large and combinations of these attributes may be used. Examples taken from various maps published are given for some explanations. See references at end for additional explanations of geologic features and common map symbology.

Discussion about Faults

Faults are a unique type of geologic feature that are one of the more difficult features to handle in GIS databases and thus commonly cause confusion. They can be created and attributed in two basic ways: (1) break faults into segments in the arc coverage and attribute each part of the fault separately (dotted, dashed, solid, solid doubling as a contact, queried, etc.), allowing each segment to form the boundary of a polygon as necessary; or (2) convert all faults on the arc coverage to “contacts” and treat them only as polygon boundaries on that coverage; then duplicate them on a separate coverage where they are attributed by type. The UGS requires that the first method be followed. The second method creates too many problems with shifted arcs and “sliver” polygons formed between duplicated fault segments as edits are made and arcs are shifted around slightly during production.

ARC - POLYGONS

Linear geologic features – some may also form boundaries of polygons.

Type – major categories of arcs and arc-polygons.

contact	boundary between mappable geologic units; usually selected at a prominent change in rock type.
fault	fracture in rock or sediment with discernable offset between opposite sides.
marker bed	prominent bed that map author has chosen to show on the map
marker bed/contact	marker bed that also serves as a contact.
moraine	mound or linear ridge of glacial debris, generally till, deposited by a glacier.
scarp	steep slope or cliff at the upper end of a landslide or other geomorphic feature (head, main, landslide, etc. is given in subtype).
joint	fracture in rock with no discernable offset between opposite sides.
lineament	linear feature of known or unknown origin visible on ground, map, or air photos.
scarp	steep slope or increase in slope of ground surface created by movement on a fault or landslide.
closed depression	naturally created small or large basin or low area in ground surface.
shoreline	linear feature created by still-stand of a lake; may be erosional or constructional.
water boundary	mapped edge of a lake, river, pond, or other water body.
snowfield boundary	mapped edge of a permanent or semi-permanent snowfield.
map boundary	defined edge or limit of the map.
dike	cross-cutting intruded body of rock, commonly igneous magma, but sedimentary dikes also occur in some areas.
vein	fracture containing mineralized material; may be discordant or concordant; quartz and calcite are common vein materials.

Subtype – primary characteristics or defining features of the types listed above; terms explained below may apply to one, many, or all of the types listed above. Feature may be shape, type of movement (faults), age, or some other characteristic deemed important by map author.

normal	type of fault that is typically moderate- to high-angle and the hanging wall (upper block) appears to display extensional offset.
reverse	type of fault that is typically moderate- to high-angle and the hanging wall (upper block) appears to display compressional offset.
thrust	type of reverse fault that is typically at a low-angle, and is commonly, but not always parallel or at low angle to bedding planes.
detachment	type of low-angle extensional fault; sometimes referred to as a “gravity fault”.
strike-slip	fault on which movement was parallel to the fault’s strike (opposite side moved left or right in reference to the observer).
attenuation	extensional fault or fault zone that thins strata.

dextral	strike-slip fault in which opposite side moved to right (right-lateral).
sinistral	strike-slip fault in which opposite side moved to left (left-lateral).
Bonneville	highest of four major named shorelines of Latest Pleistocene Lake Bonneville.
Provo	second highest of four major named shorelines of Latest Pleistocene Lake Bonneville; formed after Bonneville flood.
Stansbury	prominent intermediate named shoreline of Latest Pleistocene Lake Bonneville.
Gilbert	prominent low-level named shoreline of Latest Pleistocene Lake Bonneville.
Gunnison	a major Late Pleistocene lake.
regressive	formed during a lake regression (generally applied to shorelines).
transgressive	formed during a lake transgression (generally applied to shorelines).
intermediate	as applied to shorelines, any local prominent shoreline.
unknown	insufficient geologic data to determine.
Pinedale	major Late Pleistocene ice age that peaked about 16,000 to 23,000 years ago (C^{14} yrs b.p.) (Smiths Fork and other terms are used locally in Utah).
Bull Lake	major Late Pleistocene ice age peaking about 140,000 to 160,000 years ago (Blacks Fork and other terms are used locally in Utah).
Pine Valley	local prominent glacial episode in Late Pleistocene.
other	many other terms may be added by map author.

Modifier – additional information that further clarifies the types/features listed above; may be information such as how accurately the feature is located or understood by the map author, the type of data that was used to identify or interpret the feature, etc.

well located	position of contact, fault, or other feature is well constrained by geologic data.
approximately located	position of contact, fault, or other feature is not well constrained by geologic data.
concealed	contact, fault, or other feature is concealed by overlying deposits or water.
queried	existence of feature is questionable.
queried concealed	existence of concealed feature is questionable.
queried approximately located	position of questionable feature not well constrained.
gradational	generally a mapped contact that is placed in a broad transitional interval between different formations or lithologies.
scratch	contact or boundary between different map units based on some criteria selected by map author that is not typically used in formation and member contacts; may be a nomenclature change, a boundary across which definitions of map units changes, or a boundary separating units that are lumped together differently; commonly discordant to bedding; for example, may separate a map unit in which members of a formation are combined together from map units of the formation in which members are mapped separately.
inferred	contact, fault, or other feature inferred to be present based on circumstantial data.
geophysical	feature, usually a fault, recognized based on geophysical evidence.
other	many other terms may be added by map author.

Notes – any additional information added to attributes to help explain types/features listed above. Includes but not limited to text that appears on map. Brackets [] indicate that some defining information is to be entered, such as the geologic age of a feature.

low-angle	shallow dipping surface or lineament.
high-angle	steeply dipping surface or lineament.
steeply dipping	understood.
historic	understood.
prehistoric	understood.
[geol. age]	
[strike and dip of plane (such as a fault) - not bedding strike and dip]	
[trend and plunge]	
[key comment]	
[line name - such as name of fault]	
[dikey vein name]	
[marker bed name]	

other – many other terms may be added by map author.

Decorated lines, that is, lines with some sort of symbol on one side of the line, such as teeth on thrust faults, require that the lines be digitized in a fixed direction relative to the decorations so the decoration will plot on the side determined by the map author as shown on the source map. For example, teeth on faults are placed on the hanging wall (over-riding or upper plate or block).

Arcs bounding “overlayed” or “superimposed” polygons that overprint maps units are placed in separate coverages if the basic geologic unit is also recognized (examples of overlayed polygons are shear zones, alteration zones, and mineralized zones). For example, if part of the Isom Tuff is silicified, it can be shown on the map as Ti (Tertiary Isom Tuff) with a pattern over the silicified portion (si). The patterned area is placed in a separate polygon coverage and the bounding arcs are placed in a separate arc coverage. This differs from the way stacked units are handled. Stacked units are map units in which the map author shows a thin mantle of one map unit over another map unit. For example: Qal/Ts indicates a thin mantle of Quaternary alluvial deposits over a Tertiary unit. For GIS purposes, each stacked unit variation is treated as a unique unit with unique attributes in the map unit polygon coverage.

Polygon Names and Symbols – generally the formation, member or other map unit name, and a shorthand symbol for each.

Unit Symbol – a shorthand abbreviation of the full map unit age and name. Must be unique for every distinct map unit. The capital letter(s) always denote the geologic age (generally at period/system level) of the unit following international guidelines (see age discussion below). The small letters denote the name and/or relative age of the unit. The unit symbol is used as the label on the map, cross sections, and illustrations, and is defined in an explanation accompanying the map.

Examples:

Qa
Qap₃
QTms
Qaf/Qls
Qaf/Qls/Ct
Tgda
TRcp
PIPo
IPe
SOLf
Zmf
Xfhs

Unit Name – unique name for each map unit. Some are formally defined, indicated by upper-case letters. Some are informal terms assigned by the map author, indicated by lower-case letters.

Examples

alluvial deposits
level 3 alluvial pediment-mantle deposit
old mass-movement landslide deposit
alluvial fan deposits over lacustrine sand
alluvial fan deposits over lacustrine sand over Tintic Quartzite
Tongue A of Douglas Creek Member of Green River Formation
Petrified Forest Member of Chinle Formation
Oquirrh Formation
Ely Limestone
Laketown and Fishhaven Dolomites, undivided
Mineral Fork Formation
hornblende-biotite schist unit of Farmington Canyon Complex

(possible matching unit symbol)

Qa
Qap₃
QTms
Qaf/Qls
Qaf/Qls/Ct
Tgda
TRcp
PIPo
IPe
SOLf
Zmf
Xfhs

Age – age or age range of map unit as assigned by map author. Generally stated as accurately and precisely as evidence from geologic setting, fossils, or laboratory analyses allow (age designation in unit symbol [for example Q or IPM] is more generalized than age given here). Some examples are given below, with the letters that would be used in the unit symbol given in parentheses. See most geology textbooks or several web sites such as the U.S. Geological Survey web site at (<http://geology.er.usgs.gov/paleo/geotime.shtml>) or the Berkely University website at

(<http://www.ucmp.berkeley.edu/help/timeform.html>) for tables showing numeric ranges of geologic age terms.

Examples:

Quaternary (Q)

Quaternary-Tertiary (QT)

Pennsylvanian-Mississippian (IPM)

Lower Pennsylvanian-Upper Mississippian (IPM)

Devonian-Cambrian (DC)

Upper Proterozoic (Z)

Middle Proterozoic (Y)

Lower Proterozoic-Upper Archean (XW)

Holocene (Q)

Oligocene-Eocene (T) (for Tertiary)

Certain age characters have been long-used under international convention. Some are unique font symbols that do not have standard keyboard characters in most software. The following conventions are used for these special symbols: K – Cretaceous, TR or Tr – Triassic, IP – Pennsylvanian, C – Cambrian.

Precambrian subdivisions use Z,Y,X, and W. The following terms and symbols are rarely used, but may appear on a few geologic maps: IPM – Carboniferous (Pennsylvanian-Mississippian), CZ – Cenozoic, MZ – Mesozoic, PZ – Paleozoic, pC – Precambrian. Occasionally the map author may drop the Q or QT letters to save space – the age is given in the Map Explanation.

Notes – any additional information added to attributes to help explain types/features listed above. Includes, but not limited to, text that appears on map.

[geologic group to which unit belongs]

[modifications to age of unit made during GIS compilation]

ARCS

Structural Lines – lines used to depict structural features of the geology.

Type

structural contour	a contour (an imaginary horizontal line) drawn on a specified bedding surface to denote a structural surface; generally defined relative to sea level.
anticline	a fold, generally convex upward, whose core contains the stratigraphically oldest rock; in simple anticlines limbs dip away from plane of curvature.
syncline	a fold, generally concave upward, whose core contains the stratigraphically youngest strata; in simple synclines limbs dip toward plane of curvature.
monocline	a local steepening in an otherwise uniform planar dip; consists of two adjacent folds that form a step-like flexure.
monocline, anticlinal bend	generally the upper of the two folds forming a monocline; an inclined anticlinal fold in which the upper limb is low-angle and the lower limb is steep; may form a broad or tight anticline.
monocline, synclinal bend	generally the lower of the two folds forming a monocline; an inclined synclinal fold in which the upper limb is steep and the lower limb is low-angle; may form a broad or tight syncline.
antiform	any fold whose limbs close (converge) at the top; generally looks like a simple upright anticline except that bedding is overturned or relative age of involved rock is unknown.
synform	any fold whose limbs open (diverge) at the top; generally looks like a simple upright syncline except that bedding is overturned or relative age of involved rock is unknown.
dome	structural uplift or anticline-like fold in which beds dip away from culmination in all directions; may be circular or elliptical.
basin	structural downwarp or syncline-like fold in which beds dip toward depression from all directions; may be circular or elliptical.

Subtype

asymmetric	parts are unequal; in folds, tightness varies across hinge.
inverted	upside down, overturned, or opposite.
overturned	in folds, a fold in which one limb is overturned such that both limbs dip in the same direction.
antiformal	looks like a simple anticline; an antiformal syncline is a syncline (youngest strata in the core) that has been completely overturned such that it looks like an anticline.
synformal	looks like a simple syncline; a synformal anticline is an anticline (oldest strata in the core) that has been completely overturned such that it looks like a syncline.

Modifier

well located	see above.
projected	shown hypothetically how the feature would project or trend through space above the ground surface.
approximately located	see above.
concealed	see above.
queried	see above.
queried concealed	see above.
queried approximately located	see above.
elevation	elevation of feature above sea level.

Notes

plunging	fold axis is inclined or plunges into ground.
doubly plunging	fold axis is curved such that both ends are inclined.
isoclinal	in folds, a fold in which limbs are parallel.
recumbent	in folds, a fold in which axial plane is approximately horizontal.
open	in folds, a fold in which limbs form angle greater than 90 degrees.
closed	in folds, a fold in which limbs form angle less than 90 degrees.
[contour interval]	elevation or interval spacing of structural or other contours.
[bearing]	compass bearing.
[plunge]	amount, in degrees, of dip or inclination.
[name of 2 nd , 3 rd , etc contoured surface]	enter name of surface or feature.
[name of structure]	name of a fold or other structure [ex: Cane Creek anticline].

Isopleth Lines – any type of contoured data other than structural contour lines.

Type

thickness isopleth	thickness of a particular unit [ex: thickness of a coal bed].
chemical isopleth	chemical composition or content data [ex: copper oxide content of a particular unit].
depth isopleth	depth below ground surface of a plane.

Subtype

Examples of data or features that might be shown with isopleth lines:

coal
tar sand
gypsum
alteration zone
mineralization
chemical variations
[give name of mineral, product, or topic]

Modifier

well located	see above.
approximately located	see above.
queried	see above.
queried approximately located	see above.
scratch	see above.
projected	see above.
(can combine terms)	

Notes – includes, but not limited to, text that appears on map.

[age]
[datum]
[source of data]
[method]
etc.

Indicator Lines – extra lines placed on map to clarify or identify features or locations of supporting explanatory data. For example: cross-section lines, label leader lines, locations of measured sections.

Type

cross-section line	line on map showing position of a cross section included elsewhere in supporting materials; field is commonly labeled <u>xsection</u> .
label leader line	line indicating a feature or polygon identified by a label.
measured section	shows location or line of traverse where a section of strata discussed or mentioned in supporting materials was measured.

Subtype

(use if needed)

Modifier

(use if needed)

Notes

Examples:

A-A'

B-B'-B''

section 3a

measured section through Moenkopi Formation

Symbol Arcs – in some geologic map databases, the point data, such as strike and dip and foliation measurements, are treated as arcs that actually depict the symbol rather than as attributed point data. The reason is that point data in geology is commonly rotated to a specific orientation reported as degrees from true north. Different software create these rotations differently, possibly resulting in symbols with erroneous rotations that could mislead users. As a result, some data producers prefer to create a file that shows the symbols as arcs and annotations so that they can assure that the orientation of the symbol is correct. Most point data described below may be depicted this way in some databases.

POLYGONS

Overprinted Features – special spatial features that overprint or overlay standard map unit polygons. These are always placed in separate coverages. For example: breccia zones, altered zones, shear zones.

Unit Symbol – shorthand abbreviation of polygon name that generally appears on map as label.

Examples:

br (breccia zone)
fe (hematitic alteration)
si (silicified zone)
sz (shear zone)

[at discretion of map author, unit symbols may or may not be placed on map - the overprinted zone may only be identified on map by a pattern fill]

Unit Name – full name of overprinted polygon.

Examples:

breccia zone
hematitic alteration
silicified zone
shear zone

etc.

Modifier – generally describes contact or border type.

well located	see above.
approximately located	see above.
concealed	see above.
queried	see above.
queried concealed	see above.
queried approximately located	see above.
gradational	see above.
etc.	

Notes

[intensity term]

[source of data]

[age or event]

[other comments]

See note regarding superimposed units and stacked units on arc-polygon coverage 1.

POINT DATA

Geologic data that occur at a point, such as strike and dip and foliation measurements, are not attributed as point data in all UGS geologic map GIS databases. In some databases, point data symbols are depicted as vector files treated as arcs. See Symbol Arcs above.

Point-Source Features and Data – most map data collected at point locations and do not form polygons.

Type – primary explanation of point data.

strike and dip	strike is the direction of a line made by the intersection of a surface such as strata with the horizontal; dip is the inclination of planar strata or surfaces in the rock; generally depicted as a strike and dip symbol but may be shown using structural contours.
foliation	planar textural or structural fabric or features repeated through the rock; generally depicted as a strike and dip.
joint	planar fracture, crack, or parting in a rock; commonly repeated to form joint sets; generally depicted as a strike and dip; large joints are commonly mapped.
cleavage	in structural terms, a local planar fabric in a generally fine-grained rock that promotes a tendency to split along planes or parallel surfaces; produced by structural deformation; generally depicted as a strike and dip.
lineation	any linear geologic feature mapped at author's discretion; may be small or large.
adit	horizontal passage from surface into a mine; a tunnel; commonly includes gently inclined passages.
shaft	near vertical mine excavation or opening.
quarry	open excavation for stone or minerals.
prospect	small exploratory pit or excavation.
bore hole	hole drilled for exploration or extraction.
sample locality	location where a rock or sediment sample is collected for analysis; sample identifier and type generally given in attributes; results of analyses generally explained in accompanying materials, but may be placed on map; coverage may be given a name such as palynosample or fossilsample.
spring	location where naturally flowing water emerges from ground.
other	many other types may be used by map author.

Subtype – defines, explains, or augments point data (meaning of most of following terms is evident).

inclined	understood.
vertical	understood.
horizontal	understood.
overturned	understood.

fossil	understood.
geochemical	understood.
isotopic	sample collected for analysis of isotopes, usually to determine age of rock.
petroleum exploration	bore hole drilled to explore for petroleum resources.
water	bore hole drilled to extract water.
injection	bore hole drilled to inject water or other liquids.
mineral exploration	bore hole drilled to explore for mineral resources.
etc.	

Modifier

steep	understood.
moderate	understood.
shallow	understood.
field	data collected in field.
calculated	data calculated following established procedures.
from photogrammetry	data determined from aerial photographs visually or using computer algorithms.
dry	no oil or gas encountered in bore hole.
show oil	small amount of oil encountered.
show gas	small amount of gas encountered.
plugged and abandoned	bore hole has been plugged and capped and no longer in use.
oil	significant oil encountered in bore hole.
gas	significant gas encountered in bore hole.
oil and gas	significant oil and gas encountered in bore hole.
water	significant water encountered in bore hole.
saline brine	significant saline water encountered in bore hole.
approximately located	see above.
uncertain	insufficient data to determine how a mapped bore hole, prospect, or other feature was used or what resources were encountered.
from unpublished data	information came from unpublished sources available to author.

Examples of other information that might be provided:

hot
cold
reclaimed
modified
[type of marker or stand pipe found during mapping]
conodont
palynomorph
vertebrate
etc.

Rotation – some point data have specific orientations that author should record; symbol should be rotated on map accordingly.

Examples:

023

145.3

360

Note: map author should verify that symbols are rotated properly.

Dip – inclination in degrees from horizontal of a shaft, adit, bore hole, or other feature (angle is generally placed on map).

Examples:

0

12

45

68

Descriptor – includes but not limited to, text that appears on map.

Examples:

49±0.3 Ma

sample WA23

WA23

A

B

1989 [date]

50 bbl/day

0.03% Au

12% Cu

12 ft

43°

183°

[name]

[linked text comment]

etc.

Annotations – text that is not an attribute of a line, polygon, or point, and that is placed in a specific location on the map by the map author to locate and identify a specific feature. It generally does not include labels that are attributes of mapped polygon, arc, or point features. Most text that appears on the printed map is an attribute of an arc, polygon, or point, and can be placed anywhere along the line, within the polygon, or near the point symbol at the convenience of the cartographer, in which case it is not placed in this separate annotation category.

Type

Examples of text that could be included in this category:

[dune field; such as Antelope Valley Dune Field]
[breccia zone]
[disturbed area]
[fault zone]
[silicified in this area]
[name of a broadly defined geomorphic feature]

Examples of text that should **NOT** be included in this category:

Qal
TRm
marker bed
Wasatch fault
Virgin anticline
Texaco State #1-12
Chesterfield Mine
2,000 feet
5,280
TD -12,460
8 (dip or plunge angle)
WA23 (sample number)
U
D

Subtext

(use if needed)

Modifier

(use if needed)

Rotation

[enter rotation or orientation of text]

Point Symbols – symbols to show trend, plunge direction, sense of offset on faults, and other features selected by map author. Map author may create as needed. Examples are: arrows, balls and bars on faults, line decorations.

Type

plunge arrow	inclination of linear feature such as fold axis.
rake arrow	inclination of striations on a fault plane.
ball and bar	placed on down-dropped side of a high-angle normal or reverse fault.
up indicator	in certain cases a U may be used to indicate up-thrown side of a fault.
down indicator	in certain cases a D may be used to indicate down-thrown side of a fault.
left strike slip	arrow indicating left-lateral or sinistral movement on a strike-slip fault.
right strike slip	arrow indicating right-lateral or dextral movement on a strike-slip fault.
long arrow	used at discretion of mapper; defined in explanation of map.
short arrow	used at discretion of mapper; defined in explanation of map.
bold arrow	used at discretion of mapper; defined in explanation of map.
thin arrow	used at discretion of mapper; defined in explanation of map.

Following features may be shown symbolically by map author if too small or congested to map individually; see definitions above:

anticline
syncline
monocline
asymmetric anticline
asymmetric syncline
overturned anticline
overturned syncline
inverted anticline
inverted syncline
etc.

Subtype

(use if needed)

Modifier

(use if needed)

Rotation

[enter rotation of arrow, ball and bar, etc.]

Dip (or plunge) – place label near symbol

Examples:

5
8
23
74

Notes

[name of feature]
etc.

Additional Descriptions and Explanations of Geologic Features and Map Symbols

Geologic Data Subcommittee of Federal Geographic Data Committee, 2000, Public review draft – digital cartographic standard for geologic map symbolization: U.S. Geological Survey Open-File Report 95-525, variously paginated.

Neuendorf, K.K.E., Mehl, J.P.Jr., and Jackson, J.A., editors, 2005, Glossary of Geology, (fifth edition): Alexandria, Virginia, American Geological Institute, 779 p.